The Twist: a no-scale supergravity field for inflation and a thawing dark-energy tail, with pre-registered tests

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Version 5, consolidated after an independent referee review. Danny Lee Eldridge, independent researcher.

Abstract

We study one complex field in no-scale supergravity, $T = x + iy$ with $K = -3\ln(T + \bar{T})$. Its field space has curvature $-2/3$ (the $\alpha = 1$ attractor). This is the field space of the earlier "Spiral Twist": a dilaton φ plus a Barbero–Immirzi angle θ. The angle couples to the fermion axial current through the Nieh–Yan term with $c_A = \sqrt{(3/8)}$. That value holds for the canonically normalised angle, a normalisation we choose and state.

One explicit construction, TWIST-K, puts the cliff in a stabiliser field's Kähler metric. It gives a potential running plateau → cliff → tail.

The plateau predicts $n_s = 0.9642$–0.9654 and $r = 0.0035$–0.0038 with no fitted parameter. This prediction failed its own pre-registered test against current data. On Planck + ACT DR6 + SPT-3G + DESI DR2 + Pantheon+, fixing n_s at the Twist value costs $\Delta\chi^2 = 11.7$; the locked threshold was 9.

The tail is thawing quintessence with slope $\sqrt{(2/3)}$ and no dark-matter coupling. On a compressed late-time likelihood it was preferred to ΛCDM by $\Delta\chi^2 = -3.4$. That is 1.8σ locally, and 0.6–1.0σ after correcting for the number of late-time variants tried. The tail shape and the choice $\mu = 1$ were fixed after related data had been seen. On the full CMB + DESI + supernova data the same tail fits worse than ΛCDM ($\Delta\chi^2 = +1.45$). We therefore claim no evidence for it on current data.

Every late-time result assumes that ordinary matter does not feel the field (a "shield"). We could not realise that shield in any string setting we examined, so it is an effective-theory assumption.

What the paper offers:

- a single-field supergravity embedding with a derived axial coupling;

- predictions locked before the data that will judge them: r, the tail's w(z), no cosmic birefringence from this field, and the redshift shape of any shield leak;
- the negative results, reported in full.

Matter genesis, dark matter, the string-theory search and post-data variants are in a separate exploratory note.

1. Introduction and scope

Inflation, the matter–antimatter asymmetry and today's acceleration usually get three separate explanations. We ask how far one supergravity field can go toward all three. Every data comparison is scored against a rule written and hashed before it ran.

What is claimed.

1. The Twist's two-field geometry is the field space of no-scale supergravity at $\alpha = 1$. With $x = e^{\sqrt{(2/3)}\varphi}$, $K = -3 \ln(T + \bar{T})$ reproduces the metric exactly.
2. One construction (TWIST-K, Section 2) gives a plateau, a cliff and a $\sqrt{(2/3)}$ tail. Full two-field supergravity checks confirm that the stabiliser stays heavy and the angle stays non-tachyonic throughout.
3. Its inflation and late-time predictions are definite and falsifiable (Section 6).

What is not claimed.

- Evidence against ΛCDM (Section 4.3).
- A derivation of the cliff depth n, which plays the role of the cosmological constant.
- A string completion.
- A solution of the fifth-force problem beyond an assumed symmetry.
- A successful n_s prediction: it is currently disfavoured (Section 3.2).

How the work was done. Each data test has a written rule fixed by a SHA-256 hash before its numbers were computed. Rules are never edited. Deviations go in a separate file written before any rerun, and each test is reported once. Theory results are labelled computed, estimate or assumption.

The author is an independent researcher. The calculations, code and drafting were done with substantial help from an AI system (Claude, Anthropic), under the author's direction and these rules. Code, rule files and hashes are archived with this record (Section 8).

2. The model

2.1 Field space

The Twist has a dilaton φ and an angle θ with kinetic metric

(1) $ds^2 = d\varphi^2 + e^{\lambda\varphi}\, d\theta^2$, $\lambda = -2\sqrt{(2/3)}$

in reduced Planck units ($M_P = 2.435\times10^{18}$ GeV). Its curvature is $-\lambda^2/4 = -2/3$. Write $T = x + iy$ with $x = e^{\sqrt{(2/3)}\varphi}$ and $\theta = \sqrt{(3/2)}\, y$. Then $K_{T\bar{T}} = 3/(T + \bar{T})^2$ gives exactly (1).

The inflaton is Re T and the angle is Im T. This identification is the starting point of the work, not a result of it. It rests on the Barbero–Immirzi superfield of Gates, Ketov & Yunes [1].

2.2 Coupling to matter (explicit form)

With fermions present, the Nieh–Yan term couples the angle to the axial current [2, 3]. The coupling used throughout is

(2) $\mathcal{L} \supset (c_A/M_P)\, e^{\lambda\varphi/2}\, \partial_\mu\theta\, J_5{}^\mu = (c_A/M_P)\, \sqrt{(3/2)}\, (\partial_\mu y / x)\, J_5{}^\mu$, $c_A = \sqrt{(3/8)}$.

A rolling angle then acts as an axial chemical potential $\mu_5 = c_A\, v_\theta\, H$, where $v_\theta \equiv \sqrt{(3/2)}\, \dot{y}/(x\, H\, M_P)$ is the angle's canonical speed.

Two choices are made in (2):

- The current couples to the canonically normalised angle. The alternative, coupling to $\partial(y/x)$ through the mapping $1/\gamma_{BI} = y/x$, adds $-(y/x)\dot{x}/x$. That term is not small after inflation, and it was not computed.
- As a result, $c_A = \sqrt{(3/8)}$ is fixed only given this normalisation.

Only the exploratory matter mechanism uses (2). The inflation and late-time results do not.

2.3 The potential: TWIST-K

(3) $K = -3 \ln(T + \bar{T} - g\, E(u)\, S\bar{S} + \zeta\, E(u)^2\, (S\bar{S})^2)$, $W = \sqrt{3}\, m\, S\, T$,

with $u = T + \bar{T}$, $g = 1 + u/(2\mu)$ and $E(u) = \exp(2n(1 - e^{-\pi u/8}))$. Along $S = 0$ and real T this gives

(4) $V(x) = m^2 / [4(1 + x/\mu)] \cdot \exp(-2n(1 - e^{-\pi x/4}))$.

The three regions of (4):

- **Plateau.** It sits at small x, mirrored relative to the usual placement ($T \to 1/T$), and is an $\alpha = 1$ E-model.
- **Cliff.** The exponential factor ends inflation and lowers the energy by e^{-2n}.
- **Tail.** For $x \gg \mu$, $V \propto e^{-2n} e^{-\sqrt{(2/3)}\varphi}$. The cliff's leftover slope dies like $e^{-\pi x/4}$.

A full two-field supergravity calculation reproduces (4) to ten digits. Through the whole history S stays heavy: $m_S^2/H^2 \approx 30$ at the CMB point, $\sim 7\times10^4$ through the cliff and 2.6×10^3 today. The angle's mass squared stays positive, between $+4H^2$ and $+8H^2$.

The form $e^{-\pi x/4}$ is assumed. It resembles a worldsheet-instanton correction to a Kähler metric, but no string computation produces it or its depth (Section 7).

2.4 Parameters

This table is the single source for every number in the paper.

Symbol	Meaning	Value	Status
m	inflaton mass scale	$2.1\times10^{-5}\ M_P$	fixed by $A_s = 2.1\times10^{-9}$
n	cliff depth	124.6 (≈ 125)	chosen to give today's ρ_{DE}; $\Delta n \approx 0.35$ per factor 2. Without the $1/(1 + x/\mu)$ factor the same condition gives 126.9
μ	where the tail starts	1	chosen after late-time data were seen
ζ	quartic stabiliser term	3×10^5	chosen; S stable for $x > 8\times10^{-7}$
π/4	cliff exponent	—	assumed form
f_{heat}	share of the inflaton energy that becomes radiation	0.65–0.8%	derived from the reheating field's decay (companion note)
T_{RH}	reheating temperature	$5.5–5.8\times10^{14}$ GeV	from $H_{end} = 5.2\times10^{12}$ GeV and f_{heat}; instant reheating would give 1.9×10^{15} GeV

Symbol	Meaning	Value	Status
N*	e-folds at k* = 0.05 Mpc^{-1}	55.7 (instant) to 57.6 (derived, including kination)	computed
x_f, x_0	tail-field value at freeze and today	73, 89	computed at f_heat = 0.8%
Δφ	field distance from the CMB point to today	≈ 16 M_P	computed

Numerical studies of the cliff used n = 10 for speed. Every number quoted here is for n = 124.6.

3. Inflation

3.1 Predictions

The α = 1 plateau gives $n_s \approx 1 - 2/N^*$ and $r \approx 12/N^{*2}$. Both are evaluated exactly along the trajectory.

Reheating	N*	n_s	r
instant (f = 1)	55.7	0.9642	0.0038
derived (f_heat = 0.8%, with the kination phase)	57.6	0.9654	0.0035

Other observables at N* = 57.6:

- running $dn_s/d\ln k = -6.0\times10^{-4}$
- tensor tilt $n_t = -r/8 = -4.4\times10^{-4}$
- squeezed non-Gaussianity $f_{NL} \approx 0.014$
- energy scale V*^{1/4} ≈ 7.9×10^{15} GeV
- Hubble rate H* ≈ 1.5×10^{13} GeV

The loop correction from the reheating field is negligible once that field's mass saturates (companion note). The earlier +0.002 estimate is withdrawn. The angle is heavy during inflation ($m/H \gtrsim 2$), so it carries no isocurvature at leading order.

The locked TWIST-INFLATION rule froze the leading-order band r = 0.0037–0.0040. It reads:

- SUPPORTED if a measurement with σ(r) ⩽ 0.002 is consistent with that band within 2σ;
- RULED OUT if r < 0.002 or r > 0.008 at 95%.

The rule is scored as written. The derived-reheating value, 0.0035, lies just below the frozen band.

3.2 The pre-registered n_s test failed

PREREG_NS_SPA was scored on 13 Nov 2026. Its data were ACT DR6, SPT-3G, Planck low-ℓ, ACT lensing, DESI DR2 BAO, Pantheon+ and DES Y6 S_8, with the Twist's own late universe. Fixing n_s = 0.964 costs $\Delta\chi^2 = 11.7$ relative to n_s free (best fit 0.9756). The locked threshold was 9, so the reading is **INCONSISTENT**. ΛCDM at the same fixed n_s costs 8.5.

The high n_s comes mostly from combining the CMB with DESI BAO [4, 5]. This context does not soften the reading. Distance of the Twist's n_s from each data set:

Data	Distance
ACT alone	−0.3σ
Planck 2018	within 0.5σ
CMB-SPA alone (0.9693 ± 0.0029)	−1.7σ
Planck + ACT + DESI (0.9743 ± 0.0034)	2.7–3.3σ low
CMB-SPA + DESI (0.9737 ± 0.0025)	−3.7σ

The Twist's own dark energy does not remove the shift.

n_s cannot be raised inside the model. Reaching 0.973 would need $N^* \approx 74$. The routes that exist are too small or go the wrong way:

- a stiff post-inflation era adds at most +0.0007;
- second-order slow roll adds ~10^{-5};
- softening the cliff lowers n_s.

Variants built after seeing these data are in the companion note. They can be scored only on future data.

4. The late universe

4.1 The tail and its slope

After the cliff the field passes through a kination phase, freezes, and then thaws today on $V \propto e^{-2n}/(1 + x/\mu)$. The local slope is $\sqrt{(2/3)} \cdot x/(x + \mu)$.

Case	Share of $\sqrt{(2/3)}$
$\mu = 1$, $x = 73$–89 (computed at f_heat = 0.65–0.8%)	98.6–98.9%
$\mu = 1$, $x = 2$	67%
$\mu = 10^4$ (earlier choice)	0.02–3%: almost a cosmological constant

A share of 97% or more needs $x \gtrsim 30$. The choice $\mu = 1$ was made after the late-time data had been seen (Section 4.3).

The tail slope $\sqrt{(2/3)}$ equals $2/\sqrt{(6\alpha)}$ at $\alpha = 1$ [6]. It also equals the asymptotic Trans-Planckian Censorship bound $2/\sqrt{((d-1)(d-2))}$ in four dimensions [7].

At its best fit the tail gives $w = -0.899$ today, rising from -0.996 at $z = 3$. The same model gives two w_a values under two fitting conventions:

Convention	w_0	w_a
CPL fit to the field's own w, $z < 2$	−0.900	−0.143
Observer assuming ordinary dark matter, uniform in a, $z \leqslant 2$ (the locked WA-CEILING rule)	−0.905	−0.10

The curve hardly depends on the heat share: $w_0 \approx -0.90$ and $w_a \approx -0.14$ for f_heat between 0.01% and 0.8%.

4.2 Data and tests

The late universe is modelled as a scalar with $V \propto e^{-\kappa\varphi}$, optionally coupled to dark matter with strength β_{DM}. The supergravity model predicts $\kappa = \sqrt{(2/3)}$ and $\beta_{DM} = 0$, because masses in this supergravity class do not depend on the inflaton [8].

The compressed likelihood combines Pantheon+ supernovae (absolute magnitude marginalised), CMB distance priors, and BAO from either DESI DR2 or SDSS DR12/DR16. Each version is fitted three ways:

1. with DESI;
2. with no BAO, then predicting DESI and SDSS;
3. with SDSS in place of DESI.

All entries are differences from ΛCDM; negative is better.

Version	Extra parameters	ΔAIC, fit with DESI	Predicting DESI	Predicting SDSS	Fit to SDSS	Locked reading
PURE-TWIST (κ = √(2/3), β_DM = 0)	0	−3.37	−0.89	−0.41	−2.67	PREFERRED
KAPPA-ALPHA1 (κ = √(2/3), β_DM fitted)	1	−4.30	−6.72	+0.90	−3.04	not predictive
FLAG-1 (κ = 1, β_DM fitted)	1	−3.32	+10.88	+10.38	−1.20	not predictive
SPIRAL (κ, β_DM fitted)	2	−2.33	−6.27	+0.39	−3.27	not predictive
ON2 (coupling switches on late)	1	+0.53	−1.90	−0.37	−0.78	predicts, not preferred
SPIRAL0 (κ = 1, β_DM = 0)	0	+1.11	+0.62	−0.36	+1.02	not predictive
w_0w_a (for comparison)	2	−2.21	−3.87	−0.17	−3.27	—

DESI-HOLDOUT Test A defines its holdout differently. It refits FLAG-1 on non-DESI data with its own nuisance settings, and then FLAG-1 predicts DESI worse than ΛCDM by +12.5. The +10.88 in the table comes from the tournament's no-BAO leg.

4.3 How much the late-time result is worth

The PREFERRED reading rests on choices made after data were seen:

1. The original cliff failed its locked test (TAIL-F, $\Delta\chi^2 = +29.2$), and the cliff was then reshaped.
2. $\mu = 1$ was chosen knowing it gives the steep tail.
3. The PURE-TWIST rule was written after SPIRAL0 and KAPPA-ALPHA1 were known.

4. The same data were used for six tournament variants, TAIL-F, NS-PURE and two data-swap tests.

The size of the preference:

- **Local.** $\Delta\chi^2 = -3.37$ is a likelihood ratio of about 5, or 1.84σ locally (two-sided $p = 0.066$).
- **After trials.** With 6–12 variants tried, the Šidák-corrected significance is 0.6–1.0σ.
- **Held out.** The held-out gains (−0.89 and −0.41) are at noise level.
- **Likelihood.** The likelihood is compressed.
- **Full data.** On CMB-SPA + DESI DR2 + Pantheon+, the n_s-free leg of NS-SPA, PURE-TWIST fits worse than ΛCDM by $\Delta\chi^2 = +1.45$.

The tail is generic thawing quintessence with slope $\sqrt{(2/3)}$; its supergravity origin plays no role in these fits. Current data give **no evidence** for it. Its value is the prediction locked for new data (Section 6). DESI DR3 is cumulative with DR2, so it is a weaker out-of-sample test than Euclid.

4.4 Observables at a stated conditioning

At each model's own best fit, with neither refitted, PURE-TWIST differs from ΛCDM in BAO distances as follows:

z	D_M/r_d	D_H/r_d
0.30	+0.86%	−0.06%
0.51	+0.40%	−0.55%
0.71	+0.16%	−0.66%
0.93	0.00%	−0.63%
1.32	−0.12%	−0.45%
2.33	−0.16%	−0.10%

At the same best fits:

- $\sigma_8 = 0.781$ (ΛCDM 0.801) and $S_8 = 0.793$ (0.798);
- $f\sigma_8$ is 3.3% lower at $z = 0.5$;
- E_G is 4.9% higher at $z = 0.5$ (0.418 against 0.399).

About four points of that E_G shift come from the different best-fit Ω_m (0.310 against 0.298). At equal Ω_m the β_DM = 0 tail gives E_G within about 1% of ΛCDM, and the growth index stays GR-like (0.557 → 0.548).

The locked PREREG_GROWTH rule is stated at fixed Ω_m = 0.31 and fixed early amplitude. It predicts σ_8 1.5% lower and $f\sigma_8$ 3.2% lower at z = 0.3–0.5. At fixed Ω_m and H_0, H(z) is up to 1.9% higher near z = 0.5; this is a theoretical sensitivity, not a survey forecast.

Costs relative to ΛCDM:

- the CMB-inferred H_0 falls to ≈ 65.7 km s^{-1} Mpc^{-1};
- the late ISW signal is 5% stronger (A_ISW ≈ 1.05; measured 0.98 ± 0.35).

5. Consistency checks and negative results

The construction in Section 2 is what survived a sequence of failures:

Version tested	Result	What it forced
Plateau joined smoothly to a $\sqrt{(2/3)}$ tail	ε → 1/3, so inflation never ends	a cliff
Tail before the cliff	the tail inflates on its own (+7 e-folds)	tail after the cliff
Plateau at large T	the curved space drains the angle's speed by ~e^{-2} per e-fold	mirrored placement
A vacuum, F = T − ½	the angle oscillates 3–30,000 times per expansion time	no vacuum; true for any holomorphic F with a simple zero
Original cliff exp(−nT/(T + 1))	leftover slope dies like 1/x; TAIL-F $\Delta\chi^2$ = +29.2	the e^{−πx/4} cliff (post hoc)
Constant superpotential	no potential for the angle	the stabiliser construction

5.1 The fifth force (main open problem)

If Re T is today's dark energy, ordinary matter generically feels it:

- Through the Kaplunovsky–Louis anomaly terms [9], the QCD scale moves with x for every matter modular weight except 3/4.
- Sequestered matter gives $\beta_m \approx 0.41$, about 120 times the Cassini bound of 3.4×10^{-3} [24].
- Weight-3/4 matter leaves quark masses moving ($\beta_m \sim 0.015$).
- No standard screening mechanism applies to an exponential tail.
- Planck-suppressed operators of the kind that let the reheating field decay also generically give order-one couplings.

We therefore **assume a shield**: the Standard Model is neutral under both isometries of the T-disk, the scaling $T \to \lambda T$ and the shift $T \to T + ic$. The cliff breaks the shift only through $q = e^{-\pi T/4}$. That instanton part passes the quasar, Oklo, nucleosynthesis, atomic-clock and MICROSCOPE limits by a factor of 10^{19} or more.

The idea is not new. Carroll proposed that approximate symmetries could suppress quintessence couplings [10], and Higgs-dilaton cosmology is a precedent [11].

A perturbative leak of the form c_p/x^p needs $p \geq 4$ if its coefficient is of order one. Clocks require $|d \ln \alpha_{em}/d\varphi| < 1.6\times10^{-8}$ today. Any leak has a fixed redshift shape, locked in PREREG_SHIELD:

- about 78% of the change comes after $z = 1$;
- nothing new appears above $z \approx 3$;
- clock drifts are tied to quasar shifts.

Quantum stability of the tail needs every mass above ~6 meV to be T-independent (the Kolda–Lyth problem [12]). This is assumed.

5.2 Gravitino

$W = \sqrt{3}\, m\, S\, T$ vanishes at $S = 0$, so the inflaton sector does not set $m_{3/2}$. Supersymmetry breaking comes from a separate, unspecified sector. We take $m_{3/2}$ below the inflaton mass so the plateau is undisturbed.

The gravitino cannot be stable. In the high-scale-supersymmetry scenario of Dudas et al. [13], where all superpartners are heavier than the inflaton,

$\Omega_{3/2} h^2 \approx 0.11\, (0.1\ \text{EeV}/m_{3/2})^3\, (T_{RH} / 2\times10^{10}\ \text{GeV})^7$.

At $T_{RH} \approx 5.6\times10^{14}$ GeV this excludes a stable gravitino, so it is not the dark matter [14]. Version 4 said a gravitino heavier than T_{RH} escapes production; that misread [13] and is withdrawn.

What works is an estimate. With superpartners at or below $m_{3/2}$ and in the bath, the thermal yield is $Y \approx 1.5\times10^{-7}$. For $m_{3/2} = 10^{11}$–10^{14} GeV:

- the gravitino decays at $T \approx 4\times10^{6}$–10^{11} GeV, long before nucleosynthesis;
- it carries at most 0.4% of the energy when it decays;
- the lightest superpartner must also decay, which needs R-parity violation.

5.3 Field range and initial conditions

The field travels $\Delta\varphi \approx 16\ M_P$ from the CMB point to today. Treating T as a volume-type modulus, the tower stays at least 20 times above H and 76 times above T throughout. The far plateau ($x \lesssim 6\times10^{-6}$) lies outside the effective theory.

With $\zeta = 3\times10^{5}$, about 2.5–3.8 M_P of plateau beyond the CMB point lies inside the effective theory.

- Homogeneous starts from rest give at least 300 e-folds.
- Fast starts are excluded.
- Inhomogeneous starts fall in the robust plateau class, by analogy with published studies.

5.4 Cosmic birefringence

The field that rolls today is CP-even. Its CP-odd partner is heavy and sits at its minimum, and its coupling to fermion spin is derivative-only, which decouples [15]. The Twist therefore predicts no isotropic birefringence.

ACT + Planck report $\beta_{CMB} = 0.277° \pm 0.057°$ (4.9σ) [16], and a test of the Planck detector angles found no calibration inconsistency [17]. That result predates our lock and its absolute calibration is open, so it is not scored. A calibrated confirmation would falsify the single-field Twist.

6. Locked predictions

Each rule was hashed before its deciding data existed, and each is scored as written.

Rule	Prediction	Deciding data	Fails if
TWIST-INFLATION	r band 0.0037–0.0040 (computed value 0.0035–0.0038)	r with σ(r) ⩽ 0.002 (Simons Observatory, LiteBIRD, CMB-S4)	r < 0.002 or r > 0.008 at 95%
NS-SPA	$n_s = 0.964$	—	already scored: INCONSISTENT
WATCH-PURE-DR3	the frozen $\sqrt{(2/3)}$, $\beta_{DM} = 0$ tail; BAO as in Section 4.4	DESI DR3 BAO	DISFAVOURED if $\Delta\chi^2 \geqslant +2$ against frozen ΛCDM; SUPPORTED if $\leqslant -2$
WA-CEILING	effective $w_a \geqslant -0.40$ for every thawing Twist version	DESI DR3 $w_0 w_a$	$w_a + 3\sigma < -0.40$
GROWTH	GR-like growth; $f\sigma_8$ deficit at $z \approx 0.3$–0.5	DESI DR3 full shape, Euclid DR1, Rubin Y1	growth index off 0.55 by > 0.05 at > 3σ, or an $f\sigma_8$ excess at > 3σ
BIREF-NULL	no birefringence from this field	calibration-independent β_{CMB} with error ⩽ 0.10°	$\beta_{CMB} \geqslant 5\sigma$
LAB-NULLS, GW-FRICTION	no fifth force, no α_{em} drift, $c_{gw} = c$, $\Xi_0 = 1$	lab and gravitational-wave tests	a confirmed detection
SHIELD	any drift follows the locked redshift shape	thorium clocks, quasars, equivalence-principle tests	a drift with a different shape

Order of the decisive tests:

1. a calibrated birefringence measurement (any year);
2. DESI five-year results and Euclid first cosmology (2027);
3. the Simons Observatory, able to detect $r \approx 0.0036$ at 3–5σ (2028–2035);
4. thorium nuclear clocks, sensitive to $|d \ln \alpha_{em}/d\varphi| \sim 5\times10^{-12}$;
5. LiteBIRD (2030s).

7. Assumptions and limitations

Each assumption is a place the model could fail:

1. **The shield.** The Standard Model is neutral under the T-disk isometries. No string setting examined realises it (companion note).
2. **The cliff.** Its form e^{−πx/4} and its depth n ≈ 125 are chosen. The instanton sums we computed give about 10^{-4} of the required depth.
3. **Tail stability.** Every mass above ~6 meV is T-independent.
4. **Supersymmetry breaking.** It happens in an unspecified sector with m_3/2 below the inflaton mass, and R-parity is violated.
5. **The coupling.** The Nieh–Yan current couples to the canonically normalised angle (Section 2.2).
6. **Starting conditions.** The far plateau and fast starts lie outside the effective theory.

Limitations:

- The n_s prediction failed its locked test.
- The late-time preference is not significant after the trials correction.
- μ and the cliff form were fixed after data were seen.
- The tournament likelihood is compressed.
- The Trans-Planckian Censorship conjecture is in tension with the model, which we record as tension, not support. Its inflationary form would exclude the Twist's inflation scale. Its sharpened late-time form, $|V'/V| \geqslant \sqrt{2}$, would exclude every DESI-fitting exponential tail, this one included.
- The link to loop quantum gravity is motivation only. Today the angle sits at zero, and the self-dual value γ_BI = ±i is a choice [18].

8. Methods, data and reproducibility

The code, every rule file with its SHA-256 hash and lock time, the deviation logs and the result files are archived with this Zenodo record (AEGIS repository, branch unify-bg).

Late-time fits. These use AxiCLASS with a coupled-quintessence extension. It was validated against the unmodified build: background to 10^{-7}, and P(k) and σ_8 to 2×10^{-6}. Each fit uses two Nelder–Mead starts and must reproduce stored fits to 0.05 in χ^2.

Inflation. The two-field equations on the curved field space are solved in e-fold time, with Christoffel terms, using LSODA (rtol 10^{-8}–10^{-10}). N* uses the standard matching at k* = 0.05 Mpc^{-1}, with ρ_end = 1.5 V_end and g* = 106.75. The supergravity checks evaluate the full potential and mass matrix from (3).

Rules used. TOURNAMENT, KAPPA-ALPHA1, PURE-TWIST, DESI-HOLDOUT, MISSING-PIECE, TAIL-F, TWIST-INFLATION, NS-PURE, NS-SPA, CONSISTENCY, WATCH-PURE-DR3, WA-CEILING, GROWTH, BIREF-NULL, LAB-NULLS, GW-FRICTION, SHIELD.

9. Relation to prior work

Every ingredient has been published before:

Ingredient	Prior work
Immirzi field as a chiral superfield with K = −3 ln(Φ + Φ̄)	Gates, Ketov & Yunes [1]
Immirzi field coupled to fermions through Nieh–Yan	Mercuri [2]; Calcagni & Mercuri [3]
Plateau plus dark-energy tail in no-scale supergravity, tail slope $2/\mathcal{V}(6\alpha)$	Akrami et al. [8]; Dimopoulos & Owen [6]
Barbero–Immirzi scalar in inflation and dark energy	Pollari [19]; Gao et al. [20]
Immirzi field as a source of the baryon asymmetry	Mir et al. [21]

Mir et al. [21], read in full, differ from the Twist on every main point:

- They couple to the B−L current, with strength set by a Stückelberg mass.
- Their asymmetry comes from equilibrium leptogenesis.
- They have no inflation, dark energy or supergravity.
- They predict birefringence; the Twist predicts none.

The contribution here is narrow:

1. a stabiliser construction, TWIST-K, that gives the Gates–Ketov–Yunes superfield a plateau → cliff → tail potential with every mass under control;
2. a derived axial coupling, dependent on the normalisation choice;
3. a locked, falsifiable set of predictions, with the negative results reported.

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